AWANTIKA SRIVASTAVA

AI Engineer | LLMs & AWS

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PROFILE SUMMARY

AI Engineer with 2+ years of hands-on experience designing and deploying LLM-powered backend systems, AI Agents, and (RAG) pipelines. Strong expertise in Python backend development, LangChain/LangGraph-based agent orchestration, vector databases, and scalable API development using FastAPI. Experienced in Docker-based deployments, CI/CD workflows, and production-grade AI system design with authentication, role-based access control, and database integration.

CORE TECHNICAL SKILLS

- **Programming & Backend:** Python, FastAPI, Django, Flask, REST APIs, Microservices Architecture, Scalable API Design
- **Statistics & Mathematics:** EDA, Statistical Modeling, Descriptive & Inferential Statistics, Bayesian Statistics, Hypothesis Testing, Probability, A/B Testing
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, LightGBM, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).
- **Deep Learning:** ANN, CNN, RNN, LSTM, BERT, Transfer Learning, Self-Attention, multi-head Attention, Tensorflow, PyTorch, Keras
- **NLP:** Text preprocessing, Tokenization, Chunking, Sentiment Analysis, NLTK, TextBlob, SpaCy, Time-Series Forecasting, ARIMA, SARIMA
- **Generative AI & LLMs:** LLMs, RAG Architecture, AI Agents, LangChain, LangGraph, LlamaIndex, Prompt Engineering, Tool-Calling Workflows, Agent Memory Management, Multi-Agent Orchestration, OpenAI API, Azure OpenAI, Open-Source LLM Integration
- **Vector Databases & Retrieval:** FAISS, Chroma, Embeddings, Semantic Search, Hybrid Search, Elasticsearch, OpenSearch
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Paddle OCR, Video Analytics, OpenCV
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB
- **Deployment & DevOps:** Docker, CI/CD Pipelines, Git, GitHub, AWS (EC2, S3), Linux Deployment, RBAC, Authentication & Authorization
- **MLOps:** MLflow, Model Versioning, API Deployment, Performance Optimization

EXPERIENCE

AI/ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Designed and developed **AI-driven backend systems** using **Python** and **FastAPI** for **LLM-powered applications.**
- Built and maintained **AI Agents** using **LangChain** and **LangGraph** with **tool-calling workflows** and **memory mechanisms**.
- Developed and optimized **Retrieval-Augmented Generation (RAG) pipelines** with **embedding-based semantic retrieval**.
- Integrated **OpenAI APIs** and **open-source LLMs** into **scalable backend services**.
- Implemented **authentication**, **authorization**, and **Role-Based Access Control (RBAC)** in AI systems.
- Designed **scalable REST APIs** and **microservices architecture** for **production-ready AI applications**.
- Worked with **PostgreSQL**, **MongoDB**, and **DynamoDB** for structured and unstructured data storage.
- Deployed **containerized services using Docker** and managed **CI/CD pipelines**.
- Integrated **Elasticsearch / OpenSearch** for **hybrid search** and **vector-based retrieval**.
- Implemented **embedding generation pipelines** and optimized **vector indexing strategies** to improve retrieval latency and response accuracy.
- Designed **prompt engineering strategies** and **context window optimization techniques** to enhance LLM response relevance and reduce hallucinations.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed and deployed a** real-time computer vision–based ML system **for unsafe** driver behavior detection using **CNN-based SSD MobileNet** model.
- Trained and optimized models on large-scale video datasets, achieving **20–25 FPS** real-time processing with **<150 ms inference latency**.
- Implemented **end-to-end ML pipelines** for data ingestion, preprocessing, model training, evaluation, and production inference.
- Deployed optimized models using **TensorFlow Lite** on edge/production environments for continuous monitoring.
- Built a **Flask-based web dashboard** to visualize detections and automatically record **30-second event clips**, reducing manual review effort.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Developed **LSTM-based time-series forecasting** model trained on 5+ years historical stock data reducing **RMSE** by 18**%.**
- Engineered lag **features**, **rolling statistics**, and volatility indicators improving prediction stability.
- **Implemented** custom **BPTT training** loop with adaptive learning rate scheduling for **stable convergence**.
- Applied **regularization techniques** reducing overfitting gap by **20**%.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology